

Introduction to Microeconomics

Government Interventions in the Market

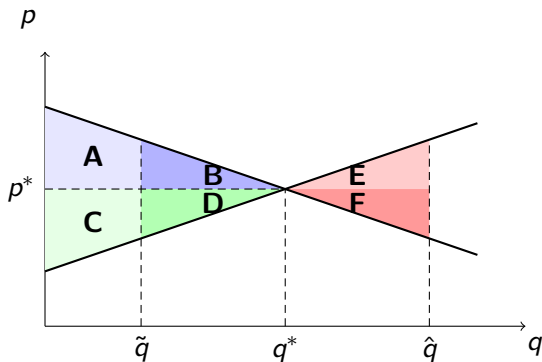
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Defining Deadweight Loss (DWL)



Definition: DWL

Deadweight Loss (DWL) is the fall in **Total Surplus** resulting from a market distortion.

1. Under-production ($\tilde{q}$):

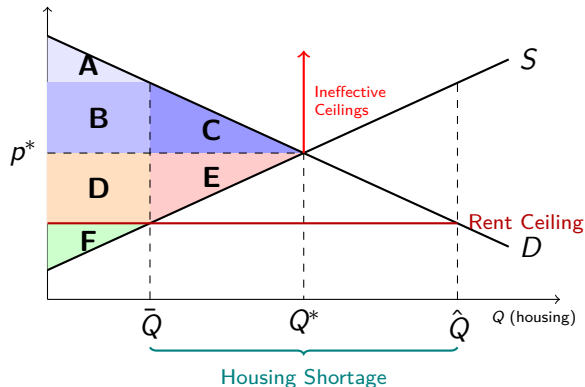
- Trades where $MB > MC$ are prevented.
- $DWL = B + D$.

2. Over-production ($\hat{q}$):

- **Surplus Destruction:** Areas $E + F$ represent a "negative" contribution to welfare.
- **Net Surplus:**
 $(A + B + C + D) - (E + F)$.

Price Ceiling: Rent Control and Surplus Distribution

p (rent price))



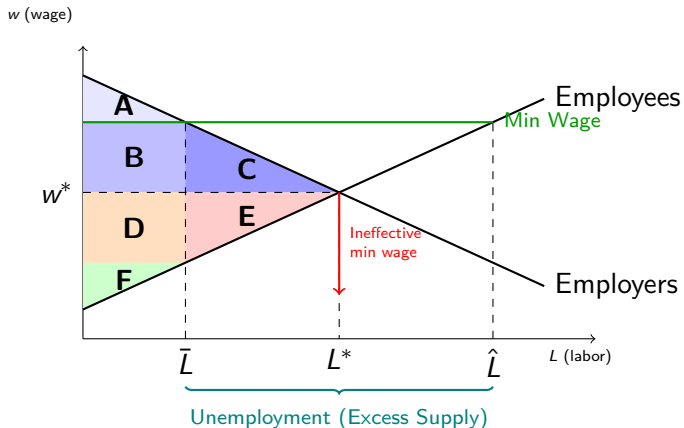
Surplus Breakdown

- **Owners (PS):** Get at least F .
- **Renters (CS):** Get at least A .
- **DWL:** $C + E$ (Loss due to shortage).

The Contested Area ($B + D$)

- **Search Costs:** If renters spend time/money finding housing, $B+D$ is lost to them.
- **Black Market:** Under-the-table payments transfer $B+D$ to owners.
- **Allocation by Luck:** If no frictions, $B+D$ stays with the renter.

Price Floor: Minimum Wage and Surplus Distribution



Surplus Breakdown

- **Employers (CS):** Get only **A**.
- **Employees (PS):** Get at least **F**.
- **DWL:** $C + E$ (Loss due to unemployment).

The Contested Area ($B + D$) *Distribution depends on market frictions:*

- **Search Costs:** If finding a job is costly, $B+D$ is lost to the worker.
- **Black Market:** Workers might "pay to work" or kick back wages, transferring $B+D$ to employers.
- **No Frictions:** $B+D$ is captured by those lucky enough to have the job.

Can Price Controls Increase Welfare?

In a perfectly competitive market, interventions **always** create inefficiency (DWL). However, the real world is rarely "perfect."

1. Correcting Market Power (Monopsony)

- **Research:** Recent work (e.g., David Card) suggests that in labor markets where employers have "monopsony power," a minimum wage can actually **increase** employment and efficiency.
- It prevents employers from artificially depressing wages to extract more profit.

2. Distributional Equity & Rent Transfers

- While total surplus may fall, the **transfer of economic rent** (areas $B+D$) from owners to low-income renters may increase social welfare if society values equity.
- If renters have a higher marginal utility of money, the utility gain for them may outweigh the utility loss for owners.

3. Search Frictions & Externalities

- If markets are plagued by high search costs or information asymmetries, price stability through regulation can sometimes reduce the costs of "panic" or predatory behavior.

Taxes

- Everything you earn and most things you buy are taxed.
- Who really pays these taxes? It turns out it is not obvious who **actually pays** and that lawmakers do not decide who ends up paying.
- **Economic vs Legal Incidence:**
 - Legal incidence: who is legally responsible to pay the tax.
 - Economic incidence: who actually bears the burden of the tax.
- **Two types:** per-unit and ad-valorem taxes.
- Taxes often lead to **under-production**, where the quantity produced ($\bar{q}$) is below the competitive equilibrium (q^*).

Subsidies

- A government payment that supports a business or market.
- Can be thought of as a "negative tax."
- Like taxes, the **economic benefit** of a subsidy does not necessarily stay with the person receiving the check.
- Subsidies often lead to **over-production**, where the quantity produced ($\hat{q}$) exceeds the competitive equilibrium (q^*).

Core Concepts

- **Definition:** Tax incidence is the division of the burden of a tax between buyers and sellers.
- **Price Impact:** When an item is taxed, its price might rise by the full amount of the tax, by a lesser amount, or not at all.
- **Who Pays?**
 - If price rises by the *full amount*, buyers pay the tax.
 - If price rises by a *lesser amount*, buyers and sellers share the burden.
 - If price does *not rise*, sellers pay the tax.

Key Insights

- **Independence from Law:** Tax incidence does **not** depend on tax law!
- The law might impose a tax on buyers or sellers, but the final economic outcome remains the same.
- **Market Forces:** The ultimate distribution of the tax burden is determined by the relative elasticities of supply and demand, not by legislative decree.

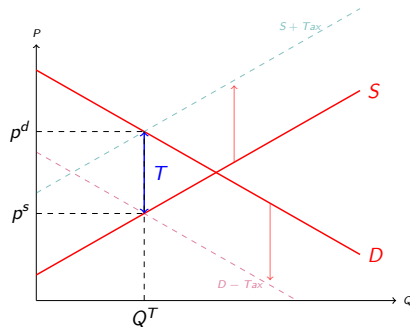
Subsidies

- Similarly, the benefit of a subsidy is shared between participants regardless of who legally receives the payment.

Taxes: The Price Wedge Approach

- **Market Equilibrium:** In the absence of restrictions, $p^E = p^d = p^s$.
- **The Tax Wedge:** A per-unit tax T introduces a gap such that $p^s \neq p^d$.
- **Key Identity:** $T = p^d - p^s$.
- **Equilibrium Condition:** The market quantity Q^T is found where:

$$Q_D(p^d) = Q_S(p^s)$$



⚠ Be careful! These auxiliary shifts don't happen.
Avoid interpreting as such!

A Crucial Distinction

A tax is a **pure movement along** the supply and demand curves because it only affects the prices faced by participants. It is **not** a shift in the underlying willingness to pay or cost of production.

The Analytics of the Tax Wedge

Assuming linear Demand and Supply:

- **Demand:** $Q_D(p^d) = \alpha - \beta p^d$
- **Supply:** $Q_S(p^s) = \gamma + \delta p^s$

A per-unit tax T is defined as $T := p^d - p^s$. To find the new equilibrium (Q^T), we solve:

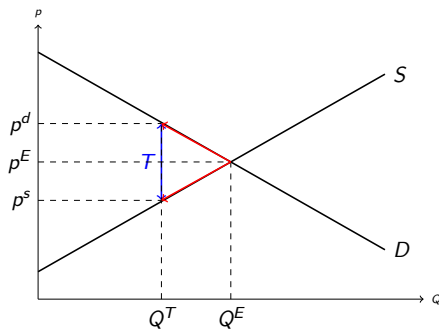
$$\gamma + \delta(p^d - T) = \alpha - \beta p^d$$

New Equilibrium Prices:

$$p^d = \frac{\alpha - \gamma + \delta T}{\beta + \delta} \quad \text{and} \quad p^s = \frac{\alpha - \gamma - \beta T}{\beta + \delta}$$

Equilibrium Quantity:

$$Q^T = \frac{\alpha\delta + \beta\gamma - \beta\delta T}{\beta + \delta}$$



Note: If $T = 0$, we recover the unrestricted equilibrium:

$$p^E = \frac{\alpha - \gamma}{\beta + \delta} \quad Q^E = \frac{\alpha\delta + \beta\gamma}{\beta + \delta}$$

Taxes: Revenue and Efficiency Loss

Government Revenue

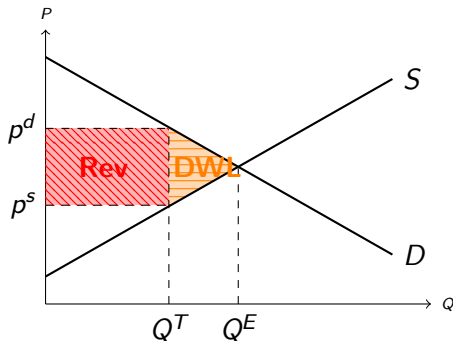
- The revenue collected is the tax per unit multiplied by the new equilibrium quantity:

$$\text{Rev}(R) = T \cdot Q^T = T \times \left(\frac{\alpha\delta + \beta\gamma - \beta\delta T}{\beta + \delta} \right)$$

Deadweight Loss (DWL)

- The loss in efficiency is the area of the DWL triangle:

$$\text{DWL} = \frac{(Q^E - Q^T) \times T}{2} = \frac{\beta\delta T^2}{2(\beta + \delta)}$$



Key Result

$\text{DWL} \propto T^2$. If $T > 0$, then $\text{DWL} > 0$ unless demand or supply are perfectly inelastic (β or δ equal to zero).

Tax Incidence and Elasticity

Defining Incidence

The tax burden shared by each side:

- **Consumer** (Tl_D): $\frac{p^d - p^E}{T} = \frac{\delta}{\beta + \delta}$
- **Supplier** (Tl_S): $\frac{p^E - p^s}{T} = \frac{\beta}{\beta + \delta}$

Properties

- $Tl_D + Tl_S = 1$
- The incidence is independent of whether the tax is legally levied on buyers or sellers.

The Elasticity Rule

Incidence can be expressed via price elasticities (multiply and divide by p/Q):

$$Tl_D = \frac{\epsilon_S^P}{|\epsilon_D^P| + \epsilon_S^P} \quad Tl_S = \frac{|\epsilon_D^P|}{|\epsilon_D^P| + \epsilon_S^P}$$

Comparative Statics

- $Tl_D > Tl_S \iff \epsilon_S^P > |\epsilon_D^P|$
- $Tl_D < Tl_S \iff \epsilon_S^P < |\epsilon_D^P|$
- “Inelasticity pays.” The more inelastic side of the market bears more of the tax.

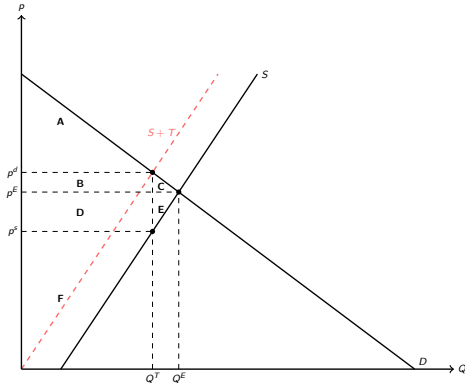
Boundary Cases

- $\epsilon_S^P = 0 \implies Tl_D = 0$.
Perfectly Inelastic Supply $\Rightarrow$ Producers absorb the whole tax burden.
- $|\epsilon_D^P| = 0 \implies Tl_S = 0$.
Perfectly Inelastic Demand $\Rightarrow$ Consumers absorb the whole tax burden.

Numerical Application: Tax Incidence and Welfare

Given: $Q_D = 100 - 2p^d$, $Q_S = 10 + p^s$, $T = 10$

- **No Tax:** $100 - 2p = 10 + p \implies p^E = 30, Q^E = 40$
- **With Tax:** $p^d = p^s + 10$
 - $100 - 2(p^s + 10) = 10 + p^s \implies p^s = 23.33$
 - $p^d = 33.33, Q^T = 33.33$



Tax Collection Summary

Tax Paid by	Area	Amount
Consumers	B	110.99
Producers	D	222.34
Total Rev R	B + D	333.33

Welfare Impact

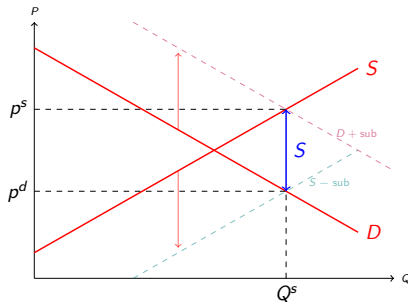
Measure	Before Tax	After Tax
CS	$A + B + C = 400$	$A = 277.81$
PS	$D + E + F = 750$	$F = 505.47$
SS	$A + B + C + D + E + F = 1150$	$A + F = 783.28$

- Note: $SS_{pre} = SS_{post} + R + DWL$
- **Deadweight Loss (DWL):** $C + E = 33.35$
- **! $\Delta SS = R + DWL$**

Subsidies: The Price Wedge Approach

- **Market Equilibrium:** In the absence of restrictions, $p^E = p^d = p^s$.
- **The Subsidy Wedge:** A per-unit subsidy s introduces a gap such that $p^s \neq p^d$.
- **Key Identity:** $S = p^s - p^d$.
- **Equilibrium Condition:** The market quantity Q^s is found where:

$$Q_D(p^d) = Q_S(p^s)$$



⚠ Be careful! These auxiliary shifts don't happen.
Avoid interpreting as such!

A Crucial Distinction

A subsidy is a **pure movement along** the supply and demand curves. It is a "negative tax" that incentivizes trade beyond the competitive equilibrium, but it does **not** change the underlying cost of production or the consumer's valuation.

The Analytics of the Subsidy Wedge

Using the same linear system:

- **Demand:** $Q_D = \alpha - \beta p^d$
- **Supply:** $Q_S = \gamma + \delta p^s$

Given $p^s = p^d + S$, we solve:

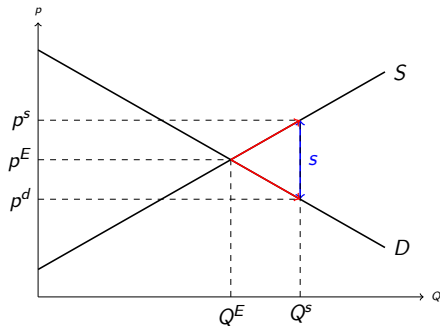
$$\gamma + \delta(p^d + S) = \alpha - \beta p^d$$

New Equilibrium Prices:

$$p^d = \frac{\alpha - \gamma - \delta S}{\beta + \delta} \quad \text{and} \quad p^s = \frac{\alpha - \gamma + \beta S}{\beta + \delta}$$

Equilibrium Quantity:

$$Q^S = \frac{\alpha\delta + \beta\gamma + \beta\delta S}{\beta + \delta}$$



Note: If $S = 0$, we recover the unrestricted equilibrium:

$$p^E = \frac{\alpha - \gamma}{\beta + \delta} \quad Q^E = \frac{\alpha\delta + \beta\gamma}{\beta + \delta}$$

Government Expenditure

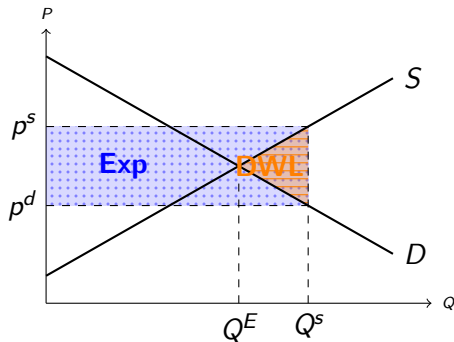
- The total cost to the government is the subsidy per unit multiplied by the new quantity:

$$\text{Exp}(X) = S \cdot Q^S = S \times \left(\frac{\alpha\delta + \beta\gamma + \beta\delta S}{\beta + \delta} \right)$$

Deadweight Loss (DWL)

- Efficiency loss from over-consumption/production (units where cost > value):

$$\text{DWL} = \frac{(Q^S - Q^E) \times S}{2} = \frac{\beta\delta S^2}{2(\beta + \delta)}$$



Key Result

$\text{DWL} \propto S^2$. If $S > 0$, then $\text{DWL} > 0$ unless demand or supply are perfectly inelastic (β or δ equal to zero).

Subsidy Incidence and Elasticity

Defining Incidence (Benefit)

The subsidy benefit shared by each side:

- **Consumer (Sl_D):** $\frac{p^E - p^d}{s} = \frac{\delta}{\beta + \delta}$
- **Supplier (Sl_S):** $\frac{p^s - p^E}{s} = \frac{\beta}{\beta + \delta}$

Properties

- $Sl_D + Sl_S = 1$
- The benefit distribution is independent of whether the subsidy is paid to buyers or sellers.

The Elasticity Rule

Benefit can be expressed via price elasticities (multiply and divide by p/Q):

$$Sl_D = \frac{\epsilon_S^P}{|\epsilon_D^P| + \epsilon_S^P} \quad Sl_S = \frac{|\epsilon_D^P|}{|\epsilon_D^P| + \epsilon_S^P}$$

Comparative Statics

- $Sl_D > Sl_S \iff \epsilon_S^P > |\epsilon_D^P|$
- $Sl_D < Sl_S \iff \epsilon_S^P < |\epsilon_D^P|$
- “Inelasticity benefits.” The more inelastic side of the market captures more of the subsidy.

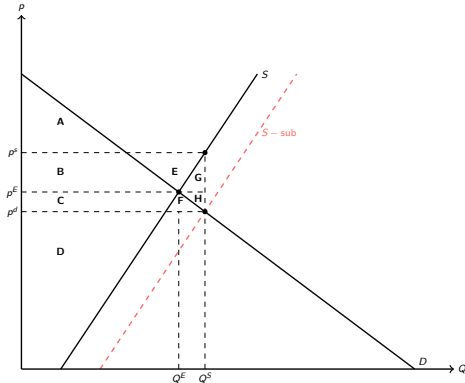
Boundary Cases

- $\epsilon_S^P = 0 \implies Sl_D = 0$.
Perfectly inelastic supply $\Rightarrow$ Suppliers capture 100% of the subsidy.
- $|\epsilon_D^P| = 0 \implies Sl_S = 0$.
Perfectly Inelastic demand $\Rightarrow$ Consumers capture 100% of the subsidy.

Numerical Application: Subsidy Incidence and Welfare

Given: $Q_D = 100 - 2p^d$, $Q_S = 10 + p^s$, $S = 10$

- **No Subsidy:** $100 - 2p = 10 + p \implies p^E = 30, Q^E = 40$
- **With Subsidy:** $p^s = p^d + 10$
 - $100 - 2p^d = 10 + (p^d + 10) \implies p^d = 26.67$
 - $p^s = 36.67, Q^S = 46.67$



Subsidy Impact Summary

Benefit to	Area	Amount
Consumers	$C + F + H$	155.411
Producers	$B + E + G$	466.7
Total Exp X	$B + C + E + F + G + H$	466.7

Welfare Impact

Measure	Before Subsidy	After Subsidy
CS	$A + B = 400$	$A + B + C + F = 544.66$
PS	$C + D = 750$	$B + E + C + D = 1039.04$
SS	$A + B + C + D = 1150$	$A + 2(B + C) + D + E + F = 1583.50$

- Note: $SS_{post} + DWL = SS_{pre} + X$
- **Deadweight Loss (DWL):** $G + H = 33.35$
- **!** $\Delta SS + DWL = X$

Agricultural Price Supports: Mechanism

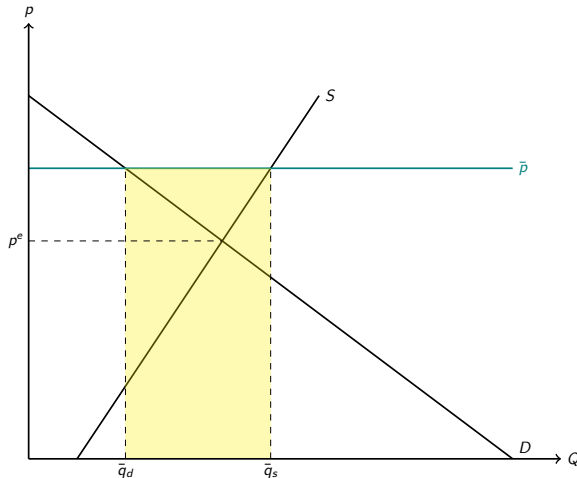
- The government sets a support price $\bar{p} > p^e$.
- To maintain this price, the government acts as the "buyer of last resort."
- **Total Supply ($\bar{q}_s$):** Determined by the supply curve at $\bar{p}$.
- **Consumer Demand ($\bar{q}_d$):** Determined by the demand curve at $\bar{p}$.
- The government purchases the **excess supply**:

$$\bar{q}_g = \bar{q}_s - \bar{q}_d$$

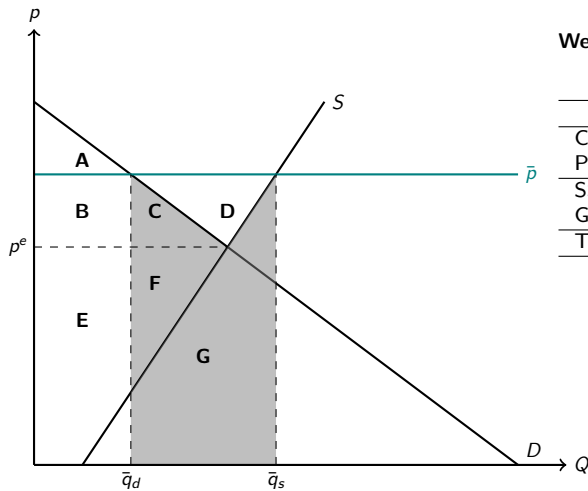
- **Total Cost (X):**

$$X = \bar{p} \times \bar{q}_g$$

- This output is typically stored, destroyed, or shipped abroad to avoid driving down domestic prices.



Welfare Analysis of Price Supports



Welfare Changes

	No Support	Support
CS	$A + B + C$	A
PS	$E + F$	$B + C + D + E + F$
SS	$A + B + C + E + F$	$A + B + C + D + E + F$
Gov (X)	0	$-C - D - F - G$
Total	$A + B + C + E + F$	$A + B + E - G$

- The change in social surplus is $\Delta SS = D$.
- Therefore

$$\Delta SS + DWL = X.$$

- As we can see, the irrevocable loss is

$$DWL = C + F + G.$$

Deficiency Payments

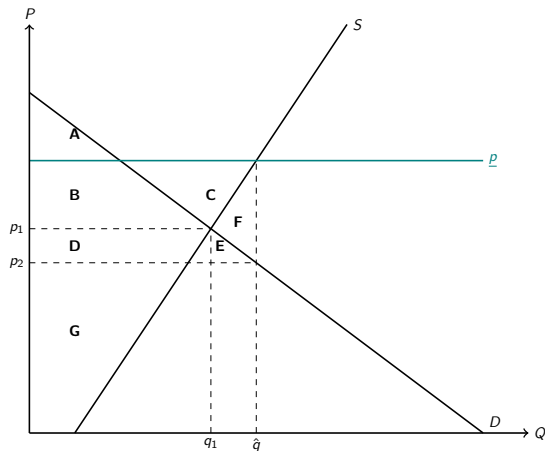
- Traditional supports led to government stockpiles of food that often spoiled.
- **Alternative:** Instead of buying output, the government sets a support price $\underline{p}$ and lets the market clear.
- Farmers produce $\hat{q}$ based on $\underline{p}$, but sell to consumers at whatever price p_2 clears the market.
- Denote the equilibrium price without any intervention as p_1 .

The Mechanism

- **Deficiency Payment:** The government pays farmers the difference: $(\underline{p} - p_2)$ per unit sold.
- Farmers receive the full support price on their entire crop.
- **Gov. Expenditure (X):**

$$X = (\underline{p} - p_2)\hat{q}$$

Welfare Analysis: Deficiency Payments



Welfare Changes

	No Support	Def. Payment
CS	$A + B$	$A + B + D + E$
PS	$D + G$	$B + C + D + G$
SS	$A + B + D + G$	$A + 2B + C + 2D + E + G$
Gov (X)	0	$-(B + C + D + E + F)$
Total	$A + B + D + G$	$A + B + D + G - F$

- **Gov. Expenditure (X):** The area $B + C + D + E + F$ represents $(\underline{p} - p_2) \times \hat{q}$.
- **Efficiency Loss:** The change in Social Surplus is exactly

$$\Delta SS = B + C + D + E$$

- Therefore, We have that

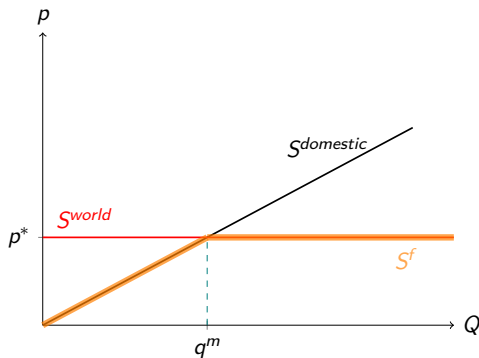
$$\Delta SS + DWL = X.$$

- Thus,

$$DWL = F.$$

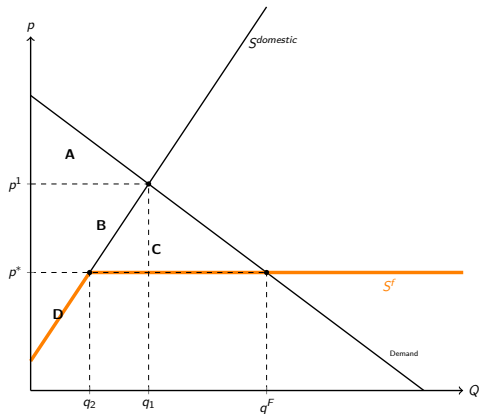
Theoretical Framework

- **Import Tariff:** A tax that applies *only* to imported goods.
- **Import Quota:** A government-imposed limit on the quantity of a good that may be imported.
- **Key Assumptions:**
 - Zero transportation costs.
 - **Small Open Economy:** The world price p^* is fixed.
 - **World Supply (S^{world}):** Perfectly elastic at p^* .



Note: S^f (orange) represents the **free-market aggregate supply**. It follows domestic supply until $P = p^*$, then becomes horizontal.

Trade Policy: Total Ban on Imports



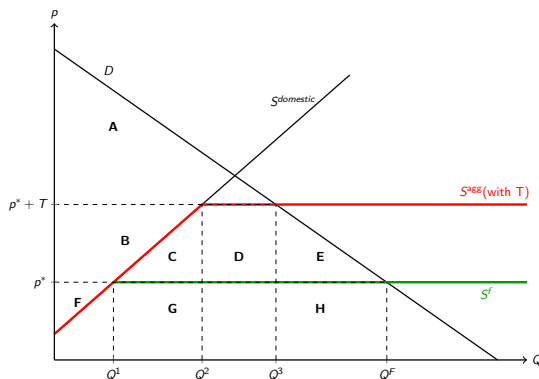
Welfare Impact of a Total Ban

Measure	Free Trade	Ban	Change
CS	$A + B + C$	A	$-(B + C)$
PS	D	$B + D$	$+B$
SS	$A + B + C + D$	$A + B + D$	$-C$

Key Result

- Area **B** is a transfer from consumers to domestic producers.
- Area **C** is the deadweight loss.
- Imports fall from $(q^F - q_2)$ to zero.

Trade Policy: Import Tariff



Key Insights

- **Government Revenues (D)**: Amount collected from import tariffs.
- **Consumption Loss (E)**: Loss from consumers who no longer buy.
- **Production Distortion (C)**: Loss from using less efficient domestic production.
- **Dead Weight Loss: $C+E$**
- **Import Reduction**: Imports fall from $(Q^F - Q^1)$ to $(Q^3 - Q^2)$.

Welfare Impact

Measure	Free Trade	Tariff	Change
CS	$A + B + C + D + E$	A	$-(B + C + D + E)$
PS	F	$B + F$	$+B$
SS	$A + B + C + D + E + F$	$A + B + F$	$-(C + D + E)$
Gov (R)	0	D	D
Total	$A + B + C + D + E + F$	$A + B + D + F$	$-(C + E)$

How a Quota Works

- An import quota is a direct **quantitative restriction** on the amount of a good that can enter the country.
- By limiting supply, it drives a wedge between the world price p^* and the domestic price p_q .
- **Equivalence**: For any tariff, there is an equivalent quota that results in the same price and quantity.

Is the Quota Binding?

- A quota $\bar{Q}$ is **binding** only if $\bar{Q} < (Q^F - Q^1)$, where $(Q^F - Q^1)$ represents free-trade imports.
- **Non-binding Quota**: If the government allows more imports than the market demands at p^* , the market remains at the free-trade equilibrium (p^*, Q^F) .
- **Binding Quota**: If $\bar{Q}$ is restricted, the domestic price must rise to p_q to reduce demand and increase domestic supply until the gap equals $\bar{Q}$.

Is the Quota Binding?

- **First Step:** Compare the quota $\bar{Q}$ to free-trade imports ($Q^F - Q^1$).
- **Non-binding:** If $\bar{Q} \geq (Q^F - Q^1)$, the quota has no effect on the market.
- **Binding:** If $\bar{Q} < (Q^F - Q^1)$, the domestic price must rise to p_q to clear the market.

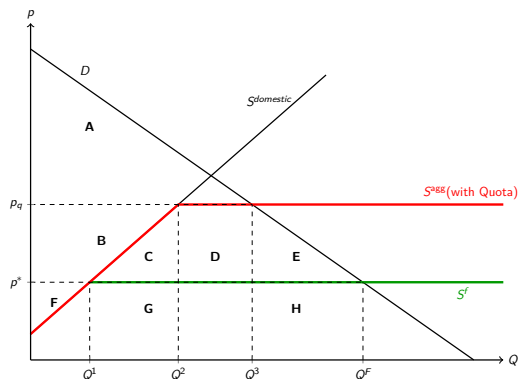
Quota Rent Allocation (D) When a quota is binding, the "rent" (Area D) is created:

- 1 **Gov. Auctions:** State captures D ; effect is identical to a tariff.
- 2 **Domestic Licenses:** Rent stays in the country as profit for importers.
- 3 **Foreign Licenses:** Foreign firms capture D , making the host country worse off.

The Efficiency Difference

While a tariff always generates domestic revenue, a quota results in a larger social loss if the quota rents are captured by foreign firms.

Import Quotas: Welfare Analysis



Key Insights

- **Quota Rent (D)**: The distribution of area D depends on how the quota is allocated.
- **Consumption Loss (E)**: Welfare loss from consumers who exit the market.
- **Production Distortion (C)**: Loss from shifting production to less efficient domestic firms.
- **Total DWL**: Either $C+E$ (if D stays domestic) or $C+D+E$ (if D is captured by foreigners).
- **Equivalence**: A quota of $\bar{Q} = Q^3 - Q^2$ leads to the same equilibrium price p_q as a tariff.

Welfare Impact of a Quota

Measure	Free Trade	Quota	Change
CS	$A + B + C + D + E$	A	$-(B + C + D + E)$
PS	F	$B + F$	$+B$
Domestic SS	$A + B + C + D + E + F$	$A + B + F (+D?)$	$-(C + E) (-D?)$
Quota Rent (D)	0	D	Depends on Allocation
Total Loss	At least $-(C + E)$; potentially $-(C + D + E)$		